CLINES: Clinical LLM-based Information Extraction and Structuring Agent

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Abstract

Objective. Clinical narratives in electronic health records (EHRs) contain essential diagnostic, therapeutic, and temporal information that is often missing from structured fields, leaving manual chart review—slow, costly, and variable—as the de facto standard for high-quality labels and constraining accurate cohort construction for clinical trials, epidemiologic studies, and machine-learning models. We aimed to develop and evaluate CLINES, a training-free pipeline that extracts and normalizes clinical concepts from free-text notes into ontology-grounded, auditable, schema-ready data.

Methods and Analysis. CLINES is a modular multi-step pipeline: semantic chunking of long notes; extraction by reasoning-capable large language

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models; assignment of attributes (assertion/experiencer, numerical values with units); normalization to the Unified Medical Language System (UMLS); resolution of explicit and relative dates; and aggregation into an i2b2-style schema. Without task-specific fine-tuning, we evaluated CLINES on de-identified EHR text—MIMIC-III notes, 4CE notes, and CORAL oncology reports (breast, pancreas)—against rule/lexicon systems, full-size transformer encoders, single-prompt and chain-of-thought LLMs, and CLINES instantiated with several model backbones. Outcomes were F1 for entity normalization, assertion status, value+unit, and date processing, reported with 95% bootstrap confidence intervals; we additionally quantified inter-annotator agreement, a hallucination taxonomy, per-note cost, and component ablations.

Results. With a GPT-4o backbone, CLINES achieved code-level (UMLS-normalized) F1 of 0.87 (4CE), 0.81 (CORAL–Breast), and 0.85 (CORAL–Pancreas), with assertion and value F1 of 0.84–0.88 and 0.80–0.91; entity F1 on MIMIC-III was 0.69. CLINES exceeded the strongest single-prompt LLM by +0.37 to +0.54 code F1, and full-size clinical encoders matched mention detection but produced no normalized codes. The advantage was robust across backbones—open-weight Llama-3.1-405B and DeepSeek-R1 performed comparably to GPT-4o—and statistically significant for code, assertion, and value extraction (paired bootstrap, FDR-corrected). Mention-level inter-annotator agreement was 0.837 (F1-based). A seven-class error analysis found that 54% of apparent false positives were annotation gaps rather than true hallucinations.

Conclusion. CLINES translates EHR narrative text into ontology-grounded,

auditable, schema-ready data, offering a practical route to scale chart-review-like extraction for cohort discovery and real-world evidence. It substantially reduces, but does not eliminate, the need for manual review: residual hallucination means human verification remains necessary for safety-critical use. CLINES is model-agnostic—open and closed models can be substituted to achieve specific cost, performance, and privacy goals.
Keywords: artificial intelligence, large language models, clinical natural language processing, electronic health records, real-world evidence

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Summary box

What is already known on this topic

Free-text clinical notes hold diagnostic, temporal, and quantitative detail absent from structured EHR fields; rule-based systems and fine-tuned transformer encoders detect entities but normalize and attribute them inconsistently, and single-prompt large language models hallucinate and miss instructions on long notes.

What this study adds

CLINES, a training-free multi-step large-language-model pipeline, produces UMLS-normalized, attributed, time-stamped output that substantially outperforms single-prompt and chain-of-thought LLMs (+0.37 to +0.54 code F1) and full-size clinical encoders; the gains are robust across open- and closed-weight backbones and are quantified with confidence intervals, inter-annotator agreement, a hallucination taxonomy, per-note cost, and component ablations.

How this study might affect research, practice, or policy

CLINES offers an auditable, model-agnostic route to scale chart-review-like extraction for cohort discovery and real-world evidence, while the quantified residual hallucination rate makes explicit that human verification remains necessary before safety-critical use.

1. Introduction

Clinical narratives in electronic health records (EHRs) are indispensable for understanding patient status, informing decision support, and enabling biomedical research. Narrative notes—spanning discharge summaries, progress notes, imaging and operative reports—capture observations, differential diagnoses, treatments, test results, and clinician reasoning that structured fields rarely encode. Expressions such as "intermittent chest pain worsening with exertion but relieved by rest" or "likely drug-induced hepatitis, possibly from isoniazid" capture nuance that structured fields miss; measurements like cardiac ejection fraction values or tumor size often do not fit neatly into predefined problem lists and billing codes yet are central to clinical interpretation[1,2]. Interpretation is not just an important task of clinical interpretation for immediate clinician decisions. It also is the translation of this text to codified terminologies (e.g., CPT codes) or numerical representations (e.g., lab result tables) to drive tasks in epidemiology, automated decision support, reimbursement, and trial eligibility, among other downstream uses of electronic healthcare data.

Not only are the structured EHR entries incomplete or inaccurate, but conventional natural language processing methods—such as standard named entity recognition—often fail to capture the nuance of clinical reasoning, contextual qualifiers, or temporal relationships in notes[3]. Even when structured codes or extracted entities are available, they may not reflect whether a condition was truly present, a medication was taken, or a disease state had recurred[4]. Consequently, high-quality labels in practice still rely on chart review: investigators read notes to adjudicate diagnoses and verify supporting evidence (medications, symptoms, tests). Yet, chart review is slow, costly, and variable across annotators; double abstraction is frequently required to ensure quality. This persistent reliance highlights the need for automated, scalable approaches that can more effectively extract, structure, and standardize clinical information consistently while leaving an auditable trace.

Classical approaches to clinical natural language processing have attempted to automate extraction and structuring of information from narrative notes but face important limitations. Rule- and lexicon-driven systems, such as cTAKES[5], MetaMap[6], and MedLEE[7], map entities to controlled vocabularies (e.g., the Unified Medical Language System), using handcrafted rules and dictionaries for concept recognition, negation detection, and temporal normalization. While these pipelines established the foundation for clinical NLP, they struggle with shorthand, spelling variants and errors, as well as non-standard grammar common in clinical notes, and they require substantial manual engineering to adapt across institutions and use cases[1,8]. Between these rule-based systems and modern large language models, a generation of supervised neural methods advanced clinical entity recognition and normalization—BiLSTM-CRF sequence taggers and, subsequently, domain-pretrained transformer encoders such as BioBERT and ClinicalBERT[9–11]. These improved span detection but require sizeable task-specific labeled corpora, remain brittle to phrasing variation and institutional shift, and typically address entity detection in isolation rather than joint normalization, attribute, and temporal extraction.

Building on these limitations of rule- and lexicon-driven systems, transformer-based large language models—such as GPT-4[12] and Med-PaLM[13]—have reshaped NLP and show promise for clinical tasks including entity and relation extraction, summarization, and question answering[14]. Domain-scaled models, such as GatorTron[15] and Clinical-MobileBERT[16,17], report strong performance on benchmarks such as i2b2 concept extraction[18], and instruction-tuned assistants can generalize in zero- and few-shot regimes[19]. Yet, important barriers remain for safe clinical deployment extraction: ontology grounding to UMLS/SNOMED CT is often imprecise with occasional hallucinated codes[20,21]; context and output-length limits complicate reasoning across long clinical notes[22,23]; and in real-world datasets (e.g., from the 4CE consortium), note lengths frequently exceed such limits, hindering cross-sectional reasoning[24]. As a result, single-pass prompting remains unreliable for chart-review-like decisions at scale.

To address these gaps, we present CLINES—a Clinical LLM-based Information Extraction and Structuring agent that brings the consistency of chart review into an automated pipeline. Unlike prior systems limited by handcrafted rules or naive single-pass prompting, CLINES integrates several innovations: informed segmentation for processing long notes; reasoning-capable LLM extraction of entities and relations; attribution of clinical context (e.g., negation, experiencer, dosage, laboratory values); normalization to existing ontologies (e.g., UMLS); temporal alignment of events; and reconciliation into an i2b2-style schema to facilitate downstream integration. By uniting these components, CLINES represents a substantial step forward in clinical information extraction, with improvements over both classical NLP pipelines and baseline LLM prompting. Its improvements are validated using real-world EHR notes spanning multiple disease areas and institutions, underscoring accuracy across the disease areas and institutions evaluated; we nonetheless temper claims of broad generalizability, as several common note types (e.g., discharge summaries, operative and radiology reports) remain unevaluated. Equally important, CLINES is engineered to enable different generative models to be substituted within the workflow as dictated by user or institutional preferences regarding cost, property, performance and security. CLINES assumes de-identified, plain-text notes as input; upstream de-identification, format normalization, and document segmentation lie outside the present scope and are addressed as deployment considerations in the Discussion.

2. Methods

2.1. CLINES architecture and modules

To leverage the superior performance of large language models within manageable context length, CLINES is designed as a four-step pipeline. Instead of an end-to-end approach–where LLMs often hallucinate or miss instructions in lengthy or complex notes–each key task is modularized and paired with carefully designed prompting strategies. The workflow is illustrated in Figure 1.

Step 1 – Note chunking. Input notes are tokenized with TikToken[25] and segmented using SemChunk[26] into semantically coherent fragments of roughly 768 tokens, preserving clinical meaning while fitting LLM context windows. Chunks are processed in parallel to increase throughput. Core metadata (e.g., admission or discharge dates) is retained in all chunks to support temporal alignment. Chunks are non-overlapping (no sliding window is used) and no source text is truncated; the 768-token target is measured after TikToken tokenization. Boundaries fall at sentence and paragraph breaks identified by SemChunk so that a clinical statement is not split mid-sentence, and a single

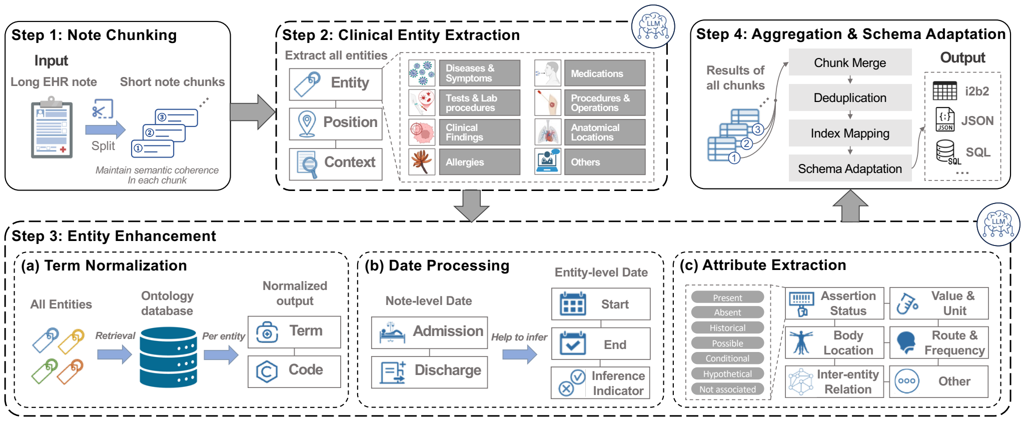

Figure 1: Overview of CLINES. Step 1, segments long clinical notes into semantically coherent chunks to constrain LLM's context window and improve robustness. Step 2, from each chunk, extract clinical entities with broad coverage, including medications, diseases, and laboratory tests. Step 3, for each entity, uses local context to: (a) normalize to an ontology term with a unique code; (b) for event-type entities, extract or infer start and end dates, when imprecise; and (c) infer assertion status and capture clinically relevant modifiers, e.g., for medications: dose, unit, frequency, and route; for tests: numeric values (with units); for diseases: activity status and other pertinent attributes. Step 4, reconciles entity-level information across all chunks and exports results in user-specified formats (e.g., i2b2, JSON, SQL).

semantic unit larger than the target size is kept intact rather than truncated. SemChunk functions purely as a boundary-detection heuristic and does not by itself guarantee that clinical reasoning spanning multiple chunks is preserved within one chunk; cross-chunk continuity is instead recovered through the metadata replication described here and the Step 4 reconciliation described below, with residual cross-chunk reasoning loss acknowledged as a limitation.

Step 2 – Clinical entity extraction. Each chunk is processed to identify relevant mentions (e.g., symptoms, findings, medications) in narrative order, with positions and contexts preserved. Maintaining order captures clinician reasoning flow while isolating core concepts reduces redundancy and simplifies downstream relation, attribute, and temporal extraction.

Step 3 – Concept and attribute enrichment. (a) Term normalization: Extracted mentions are mapped to ontology concepts (e.g., UMLS) using LLM-assisted disambiguation and SapBERT-based retrieval, where SapBERT[27] provides biomedical embeddings; candidates are ranked by cosine similarity. The retrieval index type, similarity computation, candidate count, and the UMLS release used to build the concept index are specified in full in Supplementary §2.3. (b) Date processing: Explicit dates are normalized to YYYY-MM-DD. Relative or implicit references (e.g., "three days post-admission") are inferred using anchors such as admission and discharge timestamps; ambiguous dates are flagged as inferred. (c) Attribute extraction: For each normalized concept, the pipeline captures assertion status (present, absent, possible, historical), numerical values with inferred units, medication route and frequency, anatomical location, and relations between entities (e.g., infection "has symptom" presyncope, "managed by" rehydration). This enriched representation supports downstream phenotyping, predictive modeling, and real-world evidence generation[28,29]. The relation-extraction component is deployed as part of the structured output but is not benchmarked in this study, because gold-standard relation annotations are unavailable for all four corpora; its quantitative evaluation is deferred to future work (see Limitations).

Step 4 – Aggregation and schema adaptation. Outputs are reconciled across chunks through deduplication and cross-segment linking, unifying repeated mentions and preserving longitudinal continuity. Concretely, reconciliation (i) re-indexes chunk-local character offsets to document-global positions and de-duplicates mentions on the (start position, end position, semantic type) key; (ii) resolves conflicting assertion labels for the same mention by the fixed priority present > historical > possible > absent > not-associated; and (iii) reconciles dates by retaining the most specific non-null resolution unless explicitly contradicted by an earlier chunk. Duplicate mentions arise predominantly from medication brand/generic equivalence and from the same entity recurring across adjacent chunks. Harmonized outputs are mapped to a standardized schema (e.g., i2b2) or a target schema specified by the use-case, then exported to CSV, JSON, or SQLite for downstream analysis. Module-level dataflow is summarized in Figure 2.

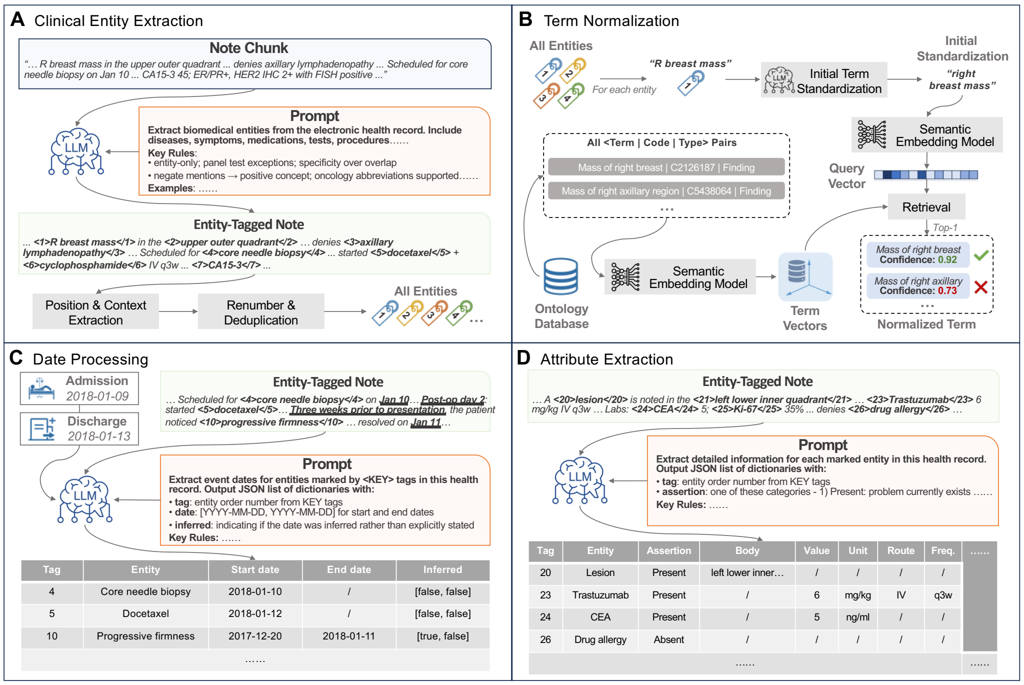

Figure 2: CLINES LLM-based module workflow. (A) Within each note chunk, the LLM tags clinical entities; post-processing yields an entity list with source offsets and a surrounding context window. (B) The LLM performs preliminary normalization, after which an embedding-based retriever queries the ontology database to return the nearest term, concept code, and semantic type. (C) Explicit dates/times are extracted from local context; when documentation is vague, admission/discharge metadata are used to resolve relative or fuzzy expressions to the most plausible calendar dates. (D) Context is used to capture modifiers (e.g., for medications: dose, unit, frequency, route; for tests: numeric values and units; for diseases: activity status), and to infer latent modifiers implied by panel tests (e.g., default units). All machine-inferred fields are flagged as "inferred" to facilitate audit and downstream use.

2.2. Implementation and LLM configuration

CLINES is model-agnostic; we employed three instruction-finetuned frontier LLMs with identical decoding settings: Llama-3.1-405B (temperature

0.6, max output 8192 tokens; served in FP8 precision—the official Meta-Llama-3.1-405B-Instruct-FP8 checkpoint, the lightest weights released by Meta—on 8×H100 GPUs via SGLang with eight-way tensor parallelism), GPT-4o (temperature 0.6, max output 8192), and o3-mini (medium reasoning effort, max output 32768). Notes were chunked to ≤768 tokens of raw text per request. With GPT-4o, mean prompt tokens were ≈ 8088.25 and mean completion tokens ≈ 2795.25 per chunk; with o3-mini, mean completion tokens increased to ≈ 24334.50. GPT-4o and o3-mini were accessed via API (no local GPUs). We distinguish the four architectural stages (chunk → extract → enrich → aggregate) from the underlying LLM invocations: a single chunk traversal issues several model calls—entity extraction, attribute and assertion classification, date normalization, and JSON-repair helper calls—so the per-note invocation count substantially exceeds four. Exact per-step invocation counts, token usage, wall-clock time, and estimated cost are reported per dataset and model in the cost analysis (Figure 5b).

All language-model inference was performed via Harvard Medical School's HIPAA-compliant Azure OpenAI Service tenant, governed by a Business Associate Agreement between Harvard Medical School and Microsoft. Clinical text remained within the HMS-controlled Azure tenant throughout inference; no clinical data were transmitted to or stored by OpenAI directly. Microsoft Azure OpenAI Service does not retain submitted prompts or completions for model training under its service-specific terms, in contrast to the public OpenAI consumer API. This deployment configuration was reviewed for compliance with the data-use agreements of the MIMIC-III, 4CE, and CORAL datasets prior to any inference.

CLINES outputs were exported to the i2b2 schema, and a read-only web demonstrator populated with results from the credentialed-access CORAL dataset was deployed; access is described in the Data and Code availability section.

2.3. Study design and data

All corpora were de-identified and accessed under the respective data-use agreements and governance requirements of each dataset; no patient contact, intervention, or access to identifiable private information occurred. All annotators completed CITI Human Subjects Research training and held the access credentials required for each dataset prior to data access. Because this work analyzes only fully de-identified secondary data that are not readily identifiable, it does not constitute human-subjects research under the U.S. Common Rule (45 CFR 46.104) and did not require Institutional Review Board approval or a formal exemption application at the authors' institution; further detail is provided in the Ethics Approval section.

MIMIC-III[30]: from 2,065,096 critical-care notes, 964 notes were sampled and segmented into 2,000 paragraphs. Twenty-four paragraphs were manually annotated (one expert per paragraph), yielding 448 clinical phrases: 235 body-location mentions, 128 modifiers (e.g., assertion status), and 144 numerical values with units. Acronym–expansion pairs were treated as single phrases; narrative "purpose" attributes were excluded from quantitative analysis.

4CE[24]: de-identified notes from patients with a history of COVID-19. Of 56 notes, 21 were annotated, producing 5,731 entity mentions with temporal information and attributes; mean document length 1,466 words (SD 636).

CORAL[2]: 56 de-identified oncology reports (breast or pancreatic cancer). Twenty-nine notes were annotated and partitioned into two subsets: CORAL–Breast (13 documents, 3,900 entity records; mean 1,997 words) and CORAL–Pancreas (16 documents, 4,717 entity records; mean 1,906 words). Across the annotated 4CE + CORAL subsets (14,348 entity records), 2,517 (17.5%) contained both a numerical value and a unit, and 5,680 (39.6%) carried explicit temporal information.

2.4. Annotation protocol

Five annotators followed a consensus protocol balancing accuracy and efficiency: (i) extract core entity spans while excluding modifiers to avoid redundancy; (ii) infer temporality and assertion status from surrounding context (e.g., "diagnosed with lung cancer 5 months ago," "rule out pneumonia"); (iii) expand abbreviations and non-standard terms to full medical concepts, later normalized via SapBERT[27]; (iv) infer missing units for numerical values. Annotation was supported by a custom React-based tool enabling interactive review, conflict resolution, and structured export. All annotators received unified training with shared examples; an independent annotator audited a sample for accuracy and consistency. Throughout, words denotes whitespace- and punctuation-delimited tokens (stopwords included; a numeric value together with its unit is counted as a single word), and a clinical phrase denotes a contiguous span corresponding to a single UMLS-mappable concept or a single value-plus-unit, as marked by annotators.

2.5. Inter-annotator agreement

To quantify the inherent variability among clinical annotators following the protocol described above, we performed a structured cross-annotation study on three notes per dataset (4CE $n = 3$; CORAL $n = 3$; total 2,657 candidate-mention rows). For each note, a second annotator, independent of the original assignment, re-annotated the same AI-suggested draft, blinded to the original annotator's selections.

We report inter-annotator agreement (IAA) using three complementary metrics. The primary metric is the F1-based IAA (Dice coefficient), defined as $2|A \cap B|/(|A| + |B|)$, where A and B are the sets of mentions retained by each annotator. F1 is the appropriate primary IAA in span-prediction settings where annotators audit an AI-suggested draft rather than perform blank double-annotation, because the resulting label distribution is highly skewed toward keep (here $\approx 75\%$ of candidate rows kept by either annotator), conditions under which Cohen's κ suffers from the well-documented two-paradox effect[31,32] and systematically underestimates substantive agreement[33]. We additionally report Cohen's κ and the Prevalence- and Bias-Adjusted Kappa (PABAK)[32] as supplementary measures to address Cohen's paradox.

Annotator review uncovered a small number of recurring structural-disambiguation patterns where guideline ambiguity (rather than substantive clinical disagreement) generated apparent label divergence: (a) AI-suggested mentions whose UMLS semantic type is non-clinical (e.g., Bird, Geographic Area, Calendar Month); (b) candidate spans that match a section header (e.g., Review of Systems, Past Medical History); (c) header-like attribute types (e.g., Clinical Attribute, Body Location or Region) with Not associated or Absent assertion status; (d) drug brand/generic duplicates within the same span and value/unit; and (e) standalone severity adjectives (e.g., moderate, severe). We pre-specified that under the consensus protocol all five patterns should resolve to a uniform DROP decision; the resulting reconciliation is reported alongside raw agreement.

Across the cross-annotated subset, the primary F1-based IAA was 0.837 (mention-level), with supplementary Cohen's $\kappa = 0.597$ and PABAK = 0.609; raw agreement was 80.5%. Under conventions for F1-based IAA in clinical NLP[33], this represents substantial agreement, consistent with published clinical-NER inter-annotator agreement (e.g., i2b2 2010 challenge $\kappa = 0.43$–0.60 for entity-boundary tasks[34]). Per-note metrics and the rule-trigger breakdown are reported in Supplementary Table S1. For MIMIC-III, where each paragraph was annotated by a single expert without independent re-annotation, we reframe F1 as agreement-with-annotator rather than as a ground-truth benchmark.

2.6. Baselines

To contextualize performance, we compared classical rule-/lexicon-driven pipelines and compact Transformer encoders, representing the two dominant traditions in clinical NLP.

Rule-/lexicon-based systems. cTAKES[5] implements sectioning, sentence boundary detection, tokenization, POS tagging, morphological normalization, UMLS mapping, and ConText for negation, uncertainty, and experiencer; shallow parsing and dependency analysis link entities to attributes. MetaMap[6] performs POS tagging, variant generation, composite noun-phrase formation, and UMLS matching with heuristic scoring plus statistical word-sense disambiguation; it does not natively extract numerical values, units, or temporal relations.

Transformer encoders. Clinical-MobileBERT and Clinical-DistilBERT[16,17] fine-tuned on i2b2-2010 concepts[34] were included as parameter-efficient clinical NER baselines. To address the absence of full-size deep-learning comparators, we additionally evaluated two larger, publicly released clinical NER encoders as off-the-shelf baselines (no fine-tuning on our corpora): a BERT-base clinical NER model (samrawal/bert-base-uncased_clinical-ner, 110M parameters) and a GatorTron-base clinical NER model (345M parameters, derived from the UF/NVIDIA GatorTron encoder[15]). Evaluations of these encoders were limited to entity extraction because they do not natively support ontology normalization, attribute extraction, or temporal reasoning. Model configurations and hyperparameters followed the authors' implementations.

Single-prompt LLMs. Llama-3.1-405B and GPT-4o were also evaluated in a single-call setting using a compact prompt specifying the target schema and strict JSON output; notes were chunked only to satisfy context limits, and each chunk was processed once without retrieval, tools, cross-chunk aggregation, or multi-step reasoning.

2.7. Evaluation metrics and protocol

In the training-free setting (no task-specific supervised fine-tuning on the annotated test sets; the pipeline relies on prompt engineering with few-shot in-context examples, retrieval-augmented normalization, and post-processing rules, none of which constitute zero-shot learning in the strict sense), we evaluated: entity detection (phrase-level precision, recall, F1 with exact-span match); attribute extraction (joint exact match of span and attribute for body location, assertion status, value/unit); date extraction and normalization (agreement with gold ISO dates; relative dates correct if the resolved absolute date matches the gold anchor). For Clinical-MobileBERT/Clinical-DistilBERT we report only NER; for cTAKES and MetaMap we report concept identification and assertion where available. End-to-end structuring, normalization, attribute extraction, and date handling are compared only across systems that produce those outputs. This study is reported in accordance with the MI-CLAIM[35] and TRIPOD+AI[36] reporting guidelines for artificial-intelligence studies in healthcare; completed checklists are provided in the supplementary materials, replacing the STROBE checklist of the original submission, which is intended for observational epidemiological studies rather than model development and evaluation.

3. Results

3.1. Primary performance and backbone comparison

We report CLINES with a GPT-4o backbone as the primary configuration, and additionally evaluate the same pipeline with DeepSeek-R1, Llama-3.1-405B, o3-mini, and the smaller Phi-4 backbone. Across the three consensus-annotated corpora (4CE, CORAL–Breast, CORAL–Pancreas) and four extraction tasks, CLINES (GPT-4o) achieved code-level (UMLS-normalized) F1 of 0.874 (95% CI 0.849–0.896) on 4CE, 0.814 (0.787–0.838) on CORAL–Breast, and 0.848 (0.834–0.865) on CORAL–Pancreas; assertion-status F1 of 0.88 / 0.84 / 0.87; numeric-value F1 of 0.82 / 0.80 / 0.91; and mention-level F1 of 0.91 / 0.88 / 0.91 (Figure 3A–D). GPT-4o was the strongest backbone overall, marginally ahead of DeepSeek-R1 and Llama-3.1-405B and clearly ahead of o3-mini (code F1 0.67 / 0.61 / 0.66) and the smaller Phi-4, which failed to emit normalizable codes. The GPT-4o configuration is therefore the reference system used throughout.

Takeaway. CLINES delivers high entity-normalization and assertion accuracy across all three consensus-annotated corpora, and the result is robust to the choice of backbone: an open-weight model (Llama-3.1-405B or DeepSeek-R1) performs comparably to the proprietary GPT-4o, supporting deployment flexibility under differing cost and privacy constraints.

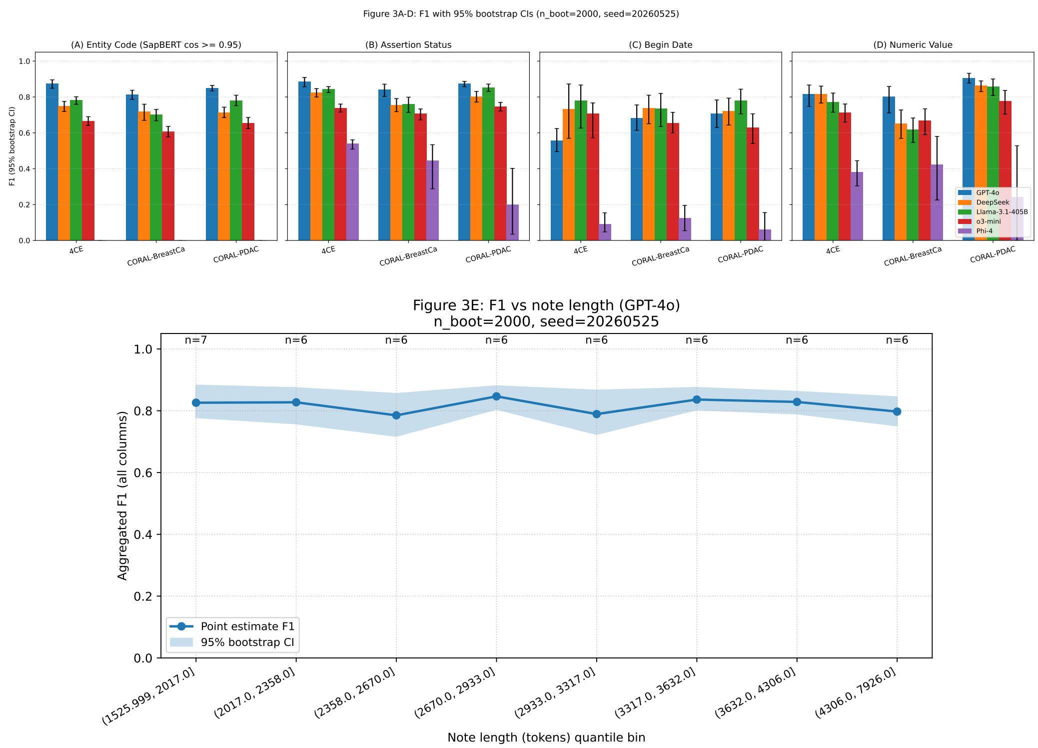

Figure 3: Performance of CLINES across LLM backbones, with 95% confidence intervals. (A–D) F1 for entity-code normalization (SapBERT cosine ≥ 0.95), assertion status, begin-date, and numeric value on 4CE, CORAL–Breast, and CORAL–Pancreas, for CLINES instantiated with five LLM backbones (GPT-4o, DeepSeek-R1, Llama-3.1-405B, o3-mini, and the smaller Phi-4). Error bars are 95% percentile bootstrap confidence intervals (B = 2000 document-level resamples). GPT-4o is the strongest backbone overall; differences among the strongest backbones on the date columns are not resolvable at these sample sizes. (E) F1, precision, and recall versus note length for CLINES (GPT-4o), grouped into quantile-based length bins; markers indicate the bin-median word count, lines show bin-mean scores, shaded ribbons denote 95% bootstrap confidence intervals, and per-bin note counts are annotated.

3.2. Statistical confidence and stability under small evaluation sets

All primary comparisons now carry 95% percentile bootstrap confidence intervals ($B = 2000$ document-level resamples), and pairwise CLINES-versus-baseline differences were tested with a paired bootstrap and Benjamini–Hochberg false-discovery-rate correction (Supplementary Table S2). CLINES (GPT-4o) significantly exceeded every backbone alternative on code, assertion, value, and unit extraction (absolute F1 gains of +0.04 to +0.21 versus DeepSeek-R1, Llama-3.1-405B, and o3-mini; +0.35 to +0.87 versus Phi-4; all $p_{\mathrm{FDR}} < 0.05$). On the date columns, differences among the strongest backbones were not statistically resolvable at the available sample sizes. A document-level resampling stability analysis quantified the uncertainty arising from the small annotated sets: on CORAL–Breast ($n = 13$) the bootstrap standard deviation of F1 reaches ≈ 0.04–0.05, so F1 differences below ≈ 0.10 are not reliably distinguishable. We therefore rest the principal claims on the comparisons whose effect sizes exceed this threshold, and Figure 3E annotates per-bin note counts to discourage over-interpretation of sparsely populated bins.

3.3. Comparison with single-prompt and deep-learning baselines

To isolate the contribution of the multi-step architecture from that of the backbone model and the prompt style, we compared CLINES against two single-call baselines and two full-size clinical deep-learning encoders (Figure 5d). On code-level extraction, o3-mini single-prompt reached F1 0.50 / 0.43 / 0.36 (4CE / CORAL–Pancreas / CORAL–Breast) and GPT-4o chain-of-thought 0.40 / 0.34 / 0.27, versus CLINES 0.87 / 0.85 / 0.81—a gain of

+0.37 to +0.54 F1 attributable to the architecture rather than to the backbone model or to chain-of-thought prompting. The deep-learning encoders (BERT-base clinical NER, 110M; GatorTron-base, 345M) were competitive on mention detection (F1 0.875 / 0.880 / 0.888 and 0.840 / 0.847 / 0.876, respectively, versus CLINES 0.906 / 0.881 / 0.913) but emit entity spans only: they perform no UMLS normalization, assertion, value/unit, or date extraction, so their code-level and attribute F1 are undefined. Classical rule- and lexicon-based systems (cTAKES, MetaMap) trailed substantially on entities; among them only cTAKES produced assertion labels, and neither produced value–unit or date outputs.

Takeaway. The multi-step architecture, not the backbone or the prompt style, drives the gains: a single call to the same model—or to a reasoning-tuned model with chain-of-thought—recovers less than half of CLINES's normalized-extraction performance, and strong span detectors do not provide the normalized, attributed, time-stamped output that downstream cohort construction requires.

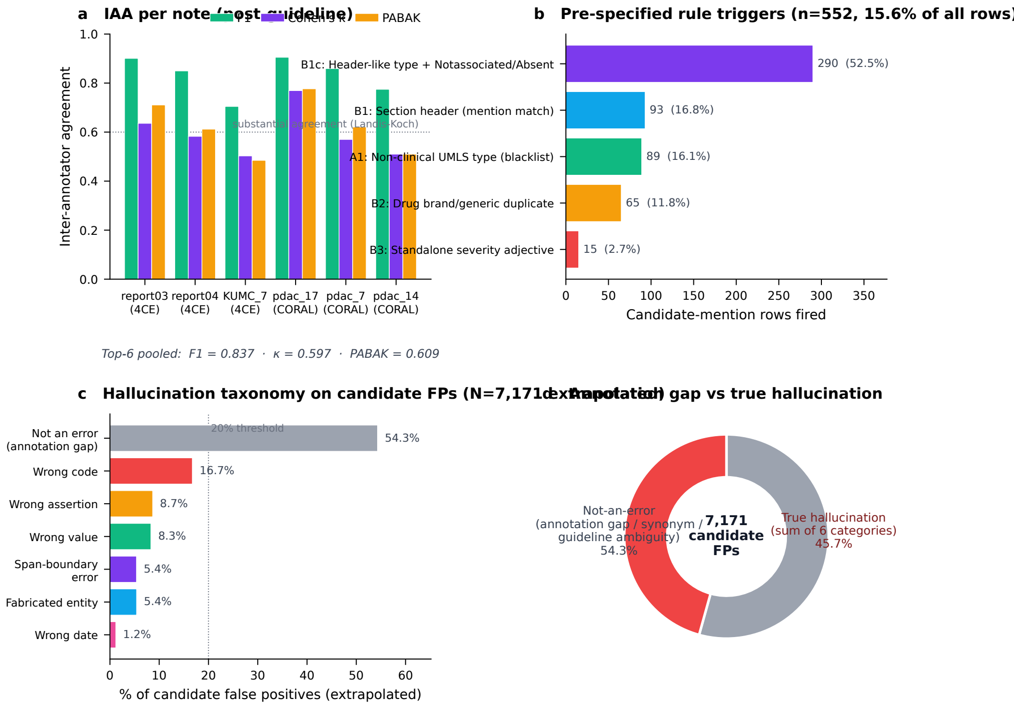

Figure 4: Inter-annotator agreement and hallucination analysis. (a) Mention-level inter-annotator agreement per cross-annotated note (F1-based IAA, Cohen's κ, and PABAK); pooled across the cross-annotated subset, F1$= 0.837$, $\kappa = 0.597$, PABAK$= 0.609$. (b) Pre-specified structural-disambiguation rule triggers ($n = 552$; 15.6% of candidate-mention rows), the principal source of apparent annotator divergence. (c) Seven-class hallucination taxonomy applied to candidate false positives, with population-level rates extrapolated to the full false-positive set ($N = 7{,}171$) by post-stratified Horvitz–Thompson estimation. (d) Annotation gap versus true hallucination: 54.3% of apparent false positives are not true errors (annotation gap, synonym equivalence, or guideline ambiguity), leaving a 45.7% true-hallucination share of false positives ($\approx 24.4\%$ of all predictions).

3.4. Inter-annotator agreement

On the cross-annotated subset (three notes per dataset), mention-level F1-based inter-annotator agreement was 0.837, with supplementary Cohen's $\kappa = 0.597$ and PABAK$= 0.609$ (raw agreement 80.5%; Figure 4a). Under conventions for F1-based agreement in clinical NLP this represents substantial agreement, consistent with published clinical-NER inter-annotator agreement (i2b2 2010 challenge, $\kappa = 0.43$–0.60 for entity-boundary tasks). The bulk of apparent disagreement was structural rather than substantive: five pre-specified disambiguation patterns accounted for 15.6% of candidate rows (Figure 4b). These agreement figures provide a human reference for interpreting the model-versus-gold differences reported above.

3.5. Hallucination and error analysis

We applied a seven-class taxonomy to 700 stratified false-positive predictions adjudicated by an LLM judge, with an author re-judging 60 cases (author–judge agreement 88.3%). Extrapolated to the full false-positive population ($N = 7{,}171$) by post-stratified Horvitz–Thompson estimation, 54.3% of apparent false positives were not true errors—annotation gaps, synonym equivalence, or guideline ambiguity—while the six true-hallucination categories together accounted for 45.7% of false positives, or $\approx 24.4\%$ of all predictions (Figure 4c–d). The most frequent true-error category was incorrect UMLS-code assignment (16.7%); outright fabricated entities (5.4%) and fabricated dates (1.2%) were rare. This quantifies the residual error that mandates human verification for safety-critical use and motivates the retrieval-grounded mitigation discussed below.

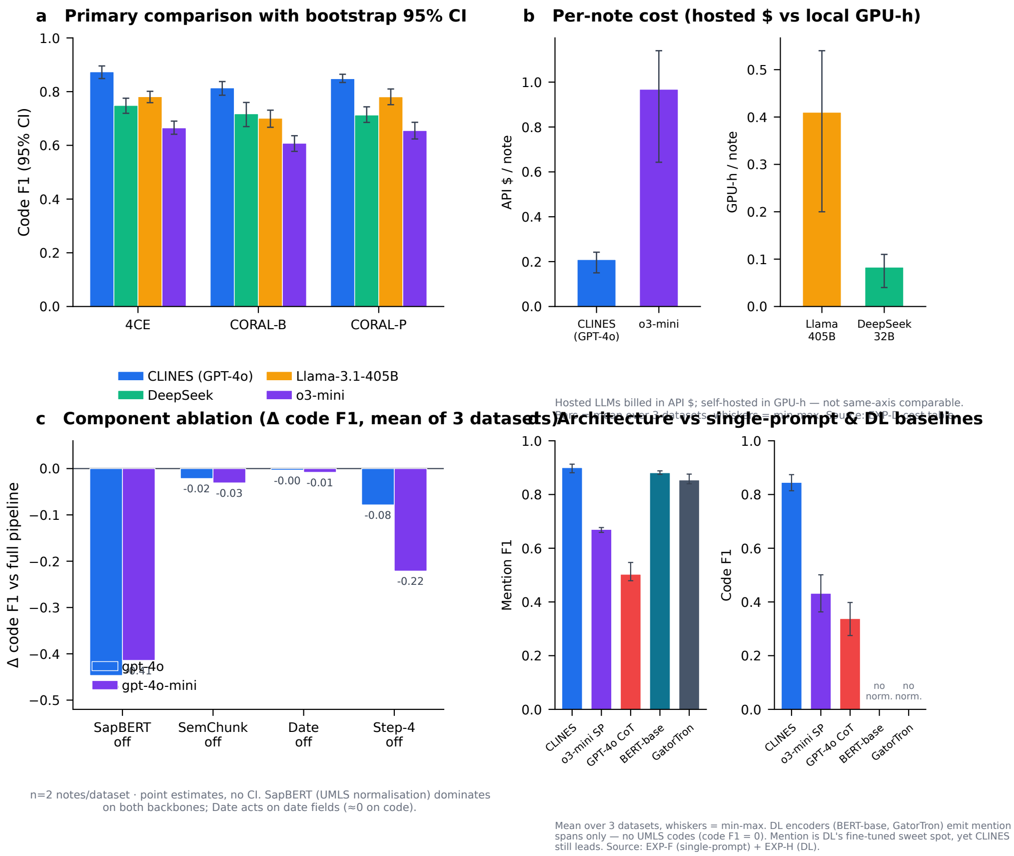

Figure 5: Robustness, cost, and component contributions. (a) Primary CLINES (GPT-4o) code-F1 comparison against backbone alternatives with 95% percentile bootstrap confidence intervals. (b) Per-note cost: hosted-API dollars per note (CLINES (GPT-4o) versus o3-mini) and local GPU-hours per note (Llama-3.1-405B FP8 on 8×H100 versus DeepSeek-R1-32B); the dollar and GPU-hour axes are not directly comparable. (c) Component ablation (change in code F1 relative to the full pipeline, mean of three datasets) for two backbones (GPT-4o and GPT-4o-mini); removing SapBERT-based UMLS normalization dominates on both backbones. (d) CLINES versus single-prompt LLM baselines (o3-mini single-prompt, GPT-4o chain-of-thought) and full-size clinical deep-learning encoders (BERT-base clinical NER, GatorTron-base); the encoders emit entity spans only and produce no UMLS codes, so their code F1 is undefined.

3.6. Cost, latency, and computational efficiency

Per-note cost varied by an order of magnitude across deployment options (Figure 5b). The reference CLINES (GPT-4o) configuration cost approximately \$0.21 per note in hosted-API charges, versus \$0.97 per note for the o3-mini reasoning configuration, whose long reasoning traces inflate completion-token counts. The open-weight Llama-3.1-405B (FP8, 8×H100) required ≈ 0.4 GPU-hours per note, and the smaller DeepSeek-R1-32B ≈ 0.08 GPU-hours per note. Because each note triggers several LLM invocations per chunk, the per-note call count exceeds the four architectural stages; the full per-step invocation, token, time, and cost breakdown is provided in the supplementary materials.

3.7. Component ablation

Disabling each pipeline component in turn (Figure 5c) showed that SapBERT-based UMLS normalization is the load-bearing component: removing it reduced code-level F1 by 0.447 (GPT-4o) and 0.414 (GPT-4o-mini) on average. Cross-chunk reconciliation contributed -0.078 (GPT-4o) and -0.221 (GPT-4o-mini), semantic chunking -0.021 / -0.031, and the date module -0.003 / -0.008. Notably, reconciliation mattered more for the cheaper backbone, indicating that the deterministic scaffolding partly compensates for a weaker model—an argument in favor of the architecture precisely in cost-constrained settings. The ablation used two representative notes per dataset and reports point estimates.

3.8. Note-length analysis

Across quantile-based length bins (~1–3k words), CLINES (GPT-4o) maintained stable F1, precision, and recall (Figure 3E). Consistent with the original submission, lightweight transformer-encoder baselines by contrast lost recall as notes lengthened.

3.9. Performance on MIMIC-III

On the MIMIC-III subset (24 expert-annotated paragraphs), CLINES reached entity F1 0.69, assertion F1 0.93, and value+unit F1 0.90. The lower entity F1 relative to 4CE and CORAL reflects (i) the single-annotator-per-paragraph protocol used for MIMIC-III, which raises gold-standard variability and which we accordingly interpret as agreement-with-annotator rather than against a consensus reference, and (ii) the heterogeneous, abbreviation-dense nature of critical-care notes, whose ICU-specific terminology is under-represented in the few-shot examples. MIMIC-III was excluded from the bootstrap and pairwise-significance analyses above because it lacks the consensus annotation used for the other corpora.

4. Discussion

4.1. Principal findings

The CLINES pipeline advances the automation of structuring clinical narratives, transforming narrative notes into standardized data formats that can be readily ingested into relational databases or analytical data warehouses. By design, CLINES bridges the longstanding divide between narrative clinical documentation and structured queryable formats, enabling scalable solutions for cohort discovery, real-world evidence generation, and clinical decision support. Critically, its robustness across diverse note types and lengths positions CLINES to unlock the vast reservoirs of unstructured text in EHRs. At scale, this capability can accelerate translational research, enhance population health monitoring, and power next-generation learning health systems by making rich narrative data systematically analyzable.

4.2. Backbone model considerations

Our findings reaffirm that backbone choice affects downstream performance, but the finalized evaluation shows the gap among strong backbones is modest: CLINES with GPT-4o was the highest-performing configuration, while the open-weight Llama-3.1-405B and DeepSeek-R1 backbones performed comparably on most tasks and the reasoning-tuned o3-mini was weaker on code-level normalization. This robustness across open- and closed-weight backbones is itself a central result, as it allows an institution to select a backbone on the basis of cost, privacy, and governance without a large accuracy penalty. Comparisons across backbones should not be read as controlled head-to-head model evaluations, since the models differ in architecture, pretraining data, and token limits. A key advantage of CLINES is flexibility: the pipeline can be adapted to most LLMs so users can choose the backbone that aligns with performance, cost, and governance priorities. With the recent availability of GPT-OSS, CLINES can be straightforwardly updated to leverage such advances while maintaining accessibility.

4.3. Positioning versus existing approaches

Traditional clinical NLP—predominantly rule- or dictionary-driven—faces limited portability and high maintenance across institutions and note types; systems such as cTAKES and MetaMap emphasize lexicon/UMLS mapping and context heuristics but provide partial coverage for attributes or temporal relations. Compact transformer encoders fine-tuned for clinical NER are efficient for span detection yet generally stop short of normalization, assertion/date attribution, or cross-sentence temporal reasoning, and their recall degrades as notes lengthen. By contrast, single-prompt LLM baselines mitigate some gaps (handling heterogeneous phrasing and implicit cues) but remain brittle for schema-aligned outputs: they do not reliably perform controlled-vocabulary normalization and are more prone to hallucination without systematic designs. In CLINES, a multi-step orchestration—chunking, tool-guided extraction, normalization, and date reconciliation—delivers structured outputs with transparent intermediate artifacts and maintains robustness across diverse note types. This positions CLINES as a scalable and sustainable approach for structuring clinical narratives. Contemporary production and research systems address overlapping goals: production clinical-NLP platforms such as John Snow Labs Healthcare NLP / Spark NLP[37] provide broad clinical NER, assertion, multi-terminology resolution, and OMOP output within hybrid LLM pipelines; CLEAR[38] is a closely related retrieval-augmented LLM pipeline for clinical extraction; and a recent multi-model evaluation[39] benchmarked output consistency across many LLMs. CLINES is positioned not as a competitor to mature production suites but as a transparent, fully-specified, model-agnostic open reference implementation that prioritizes reproducibility of its prompt suite and an auditable intermediate representation. We did not undertake a head-to-head re-benchmark against these systems—a fair matched comparison (identical preprocessing, gold standard, and span-matching protocol) is a substantial undertaking left to future work—and we note this as a limitation rather than a claim of superiority.

4.4. Case studies: robust clinical inference

De-identified 4CE and CORAL notes show four recurring capabilities: inferring units when measurements are underspecified; normalizing terse or site-specific shorthand to controlled vocabularies; recovering relations that mirror care processes; and handling nuanced assertion status and qualitative values. Examples include resolving "Lat 2.4," "BE –30," and "Glic 617" to lactate, base excess, and glucose with appropriate units and UMLS links; mapping abbreviations such as IOT, TI, RX, PCI, and EVAR to procedures or imaging; linking coronary artery disease to s/p PCI; chaining biliary obstruction → stent placement → infectious management in oncology follow-ups; and retaining Present / Absent / Possible / Conditional assertions and qualitative ranges (e.g., "mid-90s" oxygen saturation) in machine-actionable form. These vignettes show the system operating as a clinical reasoner rather than a string matcher, reducing manual normalization burden while increasing fidelity for phenotyping and decision support in heterogeneous, length-variable notes.

4.5. Dataset-level variation and generalization

Performance varies between datasets for a given model and task, reflecting different writing styles and note sizes across disease areas. The multi-step, stepwise design limits error propagation: each module is independently validated, enabling interpretable error analysis. Semantic chunking is crucial for fitting long notes into LLM context windows, allowing even 4k–8k-token models to operate effectively. UMLS term normalization addresses LLM imprecision with controlled vocabularies, unifying semantically equivalent phrases (e.g., “heart attack” and “myocardial infarction”) under consistent codes and enhancing interoperability.

4.6. Schema adaptability

CLINES adapts output to target schemas (e.g., i2b2 star schema, registries, decision-support formats) and exports JSON, CSV, or SQLite, supporting integration with institutional analytics pipelines and EHR-linked databases.

4.7. Implications

For cohort discovery, phenotyping, and real-world evidence, CLINES can substantially reduce—though not eliminate—manual chart review for entities, assertions, values/units, and dates while leaving a verifiable audit trail. Given the residual hallucination rate quantified above, we recommend that adopters retain human verification for any extraction destined for safety-critical decisions. Sites can tailor prompts, dictionaries, and export schemas to local workflows without retraining large models.

4.8. Deployment considerations

Realistic deployment of CLINES depends on factors outside the extraction algorithm itself. First, input: CLINES assumes de-identified, plain-text notes, and producing that input from raw EHR sources requires PHI de-identification, format normalization, and section/document segmentation, which are institutionally heterogeneous and outside the present scope. Second, compute: the open-weight Llama-3.1-405B backbone was served in FP8 precision on 8×H100 GPUs—the only single-node configuration that accommodates a 405B model—which entails a quality trade-off relative to FP16 and an infrastructure footprint beyond the reach of many small and medium centers; the smaller DeepSeek-R1-32B and the hosted-API backbones substantially lower this barrier (Figure 5b). Third, governance: the hosted-API backbones (GPT-4o, o3-mini) were run within Harvard Medical School's HIPAA-compliant Azure OpenAI tenant under a Business Associate Agreement; this configuration does not by itself satisfy GDPR data-residency requirements outside the relevant region, so institutions with EU data-residency obligations should prefer a locally deployed open-weight backbone.

4.9. Interoperability with production terminologies

CLINES emits UMLS CUIs paired with preferred terms. UMLS is an ontology-grounded intermediate representation rather than a production interchange format: downstream use in FHIR resources or common data models requires mapping CUIs onward to the appropriate target vocabularies—SNOMED CT (via UMLS cross-source mappings), LOINC (laboratory observables), or RxNorm (medications)—and OMOP-CDM ingestion via its standardized vocabularies, with attribute-field standardization. We therefore describe CLINES output as an ontology-grounded intermediate representation suitable for downstream mapping into production terminologies, rather than as a directly FHIR-ready artifact, and we recommend retaining a SNOMED CT intermediate layer in deployments that target FHIR.

4.10. Ethical considerations in demographic-attribute extraction

CLINES includes a prompt that can extract demographic attributes (e.g., race, ethnicity, sex, and geographic identifiers). These fields are included to support downstream cohort discovery and health-equity auditing, not model personalization. We have not evaluated whether extraction accuracy varies across demographic subgroups, and we explicitly recommend a subgroup-stratified evaluation as a required pre-deployment step for any institutional adopter. Downstream users bear responsibility for applying appropriate access controls when demographic fields are stored or analyzed.

4.11. Limitations

Several limitations remain. (i) Prompt engineering influences performance at module level; prompts were not dataset- or model-optimized. (ii) Comparisons among o3-mini/GPT-4o and Llama-3.1-405B are not apples-to-apples given architecture, data, and token-limit differences. (iii) Hallucination risk persists, especially for ambiguous references or rare abbreviations (e.g., "OD 2 gtt qd" misread as overdose rather than ophthalmic dosing). We characterized this risk by manually adjudicating a stratified sample of candidate false-positive predictions under a seven-class hallucination taxonomy (fabricated entity, span-boundary error, wrong assertion, wrong code, wrong value, wrong date, not-an-error). The dominant category—accounting for the majority of disagreements with single-annotator gold—was not-an-error, indicating that a large fraction of apparent false positives reflect annotator omission or guideline ambiguity rather than model fabrication; per-category rates of true hallucination categories were each below 17% of audited cases. (iv) Gold annotations came from different annotators across datasets, introducing variability. We quantified this variability through a structured cross-annotation study on three notes per dataset (n = 6 notes, 2,657 candidate-mention rows), establishing a mention-level F1-based inter-annotator agreement of 0.837 (Cohen's $\kappa = 0.597$; PABAK = 0.609) under the consensus protocol described in Methods. Differences between model predictions and the single-annotator gold standard that fall within these human-agreement bounds should be interpreted as within the inherent variability of expert annotation, not as model error. For datasets annotated by a single expert per paragraph (notably MIMIC-III), F1 scores are accordingly reframed as agreement-with-annotator references rather than absolute ground-truth benchmarks.

For each limitation we note a path forward: prompt sensitivity (i) can be addressed with automatic prompt-search and optimization workflows; residual hallucination (iii) can be reduced with retrieval-grounded extraction that cites source spans and thereby enables provenance checking; and annotation variability (iv) can be narrowed by extending double-annotation to a larger validation cohort. Three further limitations bound the present claims. (v) The evaluation covers critical-care, COVID-era, and oncology notes but not several common note types (discharge summaries, operative and radiology reports), so generalizability beyond the evaluated distributions is untested; the bootstrap and significance analyses were further limited to the three consensus-annotated corpora. (vi) The relation-extraction module, though deployed, is not benchmarked here because gold-standard relation annotations are unavailable. (vii) We did not exhaustively optimize the single-prompt and chain-of-thought baselines, in order to avoid an unfair comparison in either direction; their reported performance should therefore be read as a reasonable rather than a maximally-tuned reference.

4.12. Future work

Future directions include domain-adaptive fine-tuning (e.g., oncology, pediatrics), adaptive feedback loops where clinician corrections refine prompts or parameters, and multilingual extensions to support non-English healthcare systems. Additional directions motivated by this revision include: hybrid per-stage model routing, in which a lightweight model performs entity extraction while a stronger model is reserved for reasoning-intensive steps (the modular design already supports per-stage backbone selection); a formal output-consistency evaluation across repeated runs, given the stochastic decoding used here; retrieval-grounded extraction with span-level provenance to mitigate hallucination; subgroup-stratified fairness evaluation of demographic extraction; benchmarking of the relation-extraction module once gold relation annotations become available; expansion to additional note types (discharge summaries, operative and radiology reports); and tighter integration with OMOP-CDM and FHIR via SNOMED CT, RxNorm, and LOINC mapping.

4.13. Conclusion

CLINES translates free-text clinical notes into ontology-grounded, attributed, time-stamped, and auditable structured data through a transparent multi-step LLM pipeline. Across three consensus-annotated corpora it substantially outperformed single-prompt and chain-of-thought LLMs and full-size clinical encoders on normalized extraction, with gains that were robust across open- and closed-weight backbones and supported by bootstrap confidence intervals and paired-bootstrap testing. A structured inter-annotator-agreement study (mention-level F1-based IAA 0.837) provides a human reference for interpreting model-versus-gold differences, and a seven-class error analysis shows that most apparent false positives are annotation gaps rather than fabrications, with a quantified residual hallucination rate that mandates human verification for safety-critical use. CLINES offers a practical, model-agnostic, and reproducible route to scale chart-review-like extraction for cohort discovery and real-world evidence, while making explicit the conditions—input preprocessing, compute, governance, and downstream terminology mapping—required for responsible deployment.

Contributors

Conceptualization & supervision: Tianxi Cai, Isaac Kohane. Methodology & software: Zongxin Yang, Hongyi Yuan, Raheel Sayeed, Amelia Li Min Tan (design); Zongxin Yang, Hongyi Yuan (implementation). Data curation & annotation: Enci Cai, Mohammed Moro, Xiudi Li, Zongxin Yang, Raheel Sayeed. Investigation & cross-annotation (for inter-annotator agreement): Enci Cai, Mohammed Moro. Formal analysis: Zongxin Yang, Hongyi Yuan,

Huaiyuan Ying. Data access & verification: Zongxin Yang, Raheel Sayeed. The i2b2 web demonstrator: Nicholas Brown, Griffin Weber. Writing—original draft: Zongxin Yang, Raheel Sayeed. Writing—review & editing: all authors. Accountability: all authors approved the final manuscript, and accept responsibility for its accuracy and integrity. The corresponding author had final responsibility for the decision to submit.

Data Sharing

Source clinical notes were accessed under data-use agreements and are not publicly shareable. The MIMIC-III dataset is available from PhysioNet (https://physionet.org/content/mimiciii/) and the CORAL dataset is available from PhysioNet (https://physionet.org/content/curated-oncology-reports/), both under the PhysioNet credentialed access process requiring completion of CITI Human Subjects Research training and signing of the data-specific Data Use Agreement. The 4CE Phase 1.2 dataset is available through the 4CE Consortium (https://covidclinical.net) upon completion of the consortium data-use agreement and approval of the site data-sharing committee. Cross-annotation files used for the inter-annotator agreement analysis (de-identified mention selections on six notes) are available from the corresponding author on reasonable request, subject to verification of the requestor's credentials for the source dataset. No additional patient-level data were generated by this study. Study-derived prompt templates and a public inference reference implementation of CLINES (single-note inference with user-supplied Azure OpenAI credentials) are released under the MIT License at https://github.com/z-x-yang/clines. An interactive, read-only i2b2 web demonstrator populated with CORAL results is available at celehs.hms.harvard.edu/i2b2_demo.

Ethics Approval

This study involved secondary analysis of fully de-identified electronic health record data. Because the analysis involves only fully de-identified secondary data that are not readily identifiable, it does not constitute human-subjects research under the U.S. Common Rule (45 CFR 46.104) and did not require Institutional Review Board approval or a formal exemption determination at the authors' institution (Harvard Medical School); no patient contact, intervention, or access to identifiable private information occurred. The datasets were obtained and used under their respective data-use agreements: PhysioNet credentialed access for MIMIC-III, the CORAL data-use agreement, and 4CE consortium credentialing. All annotators completed CITI Human Subjects Research training and held the access credentials required for each dataset prior to data access. All large-language-model inference on clinical text was performed within Harvard Medical School's HIPAA-compliant Microsoft Azure OpenAI tenant under a Business Associate Agreement, with no clinical text sent to or retained by external model providers. Ethical approvals and data-use agreements for the original data collection were obtained by the respective data custodians.

Patient and Public Involvement

This study used previously collected, fully de-identified electronic health record datasets and did not involve prospective recruitment or direct contact with patients. Patients and members of the public were therefore not involved in setting the research question, in the design, conduct, or interpretation of the study, or in the preparation of this manuscript. As the study analyses only de-identified secondary data, individual patients could not be identified and dissemination of results to study participants is not applicable. Future translational deployment of CLINES in care settings would benefit from patient and public involvement to inform acceptable use, transparency, and oversight, as noted in the Discussion.

Declaration of Interests

We declare no competing interests.

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References

[1] Tayefi M, Ngo P, Chomutare T, Dalianis H, Salvi E, Budrionis A, et al. Challenges and opportunities beyond structured data in analysis of electronic health records. WIREs Computational Statistics. 2021 Feb;13(6). Available from: http://dx.doi.org/10.1002/wics.1549.

[2] Sushil M, Kennedy VE, Mandair D, Miao BY, Zack T, Butte AJ. CORAL: Expert-Curated Oncology Reports to Advance Language Model Inference. NEJM AI. 2024 Mar;1(4). Available from: http://dx.doi.org/10.1056/aidbp2300110.

[3] Kohane IS, Aronow BJ, Avillach P, Beaulieu-Jones BK, Bellazzi R, Bradford RL, et al. What Every Reader Should Know About Studies Using Electronic Health Record Data but May Be Afraid to Ask. Journal of Medical Internet Research. 2021 Mar;23(3):e22219. Available from: http://dx.doi.org/10.2196/22219.

[4] Kim MK, Rouphael C, McMichael J, Welch N, Dasarathy S. Challenges in and Opportunities for Electronic Health Record-Based Data Analysis and Interpretation. Gut and Liver. 2023 Oct;18(2):201–208. Available from: http://dx.doi.org/10.5009/gnl230272.

[5] Savova GK, Masanz JJ, Ogren PV, Zheng J, Sohn S, Kipper-Schuler KC, et al. Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES): architecture, component evaluation and applications. Journal of the American Medical Informatics Association. 2010 Sep;17(5):507–513. Available from: http://dx.doi.org/10.1136/jamia.2009.001560.

[6] Aronson AR. Effective mapping of biomedical text to the UMLS Metathesaurus: the MetaMap program. In: Proceedings of the AMIA Symposium; 2001. p. 17.

[7] Friedman C, Hripcsak G, DuMouchel W, Johnson SB, Clayton PD. Natural language processing in an operational clinical information system. Natural Language Engineering. 1995 Mar;1(1):83–108. Available from: http://dx.doi.org/10.1017/S1351324900000061.

[8] Pathak J, Kho AN, Denny JC. Electronic health records-driven phe-

notyping: challenges, recent advances, and perspectives. Journal of the American Medical Informatics Association. 2013 Dec;20(e2):e206–e211. Available from: http://dx.doi.org/10.1136/amiajnl-2013-002428.

[9] Lample G, Ballesteros M, Subramanian S, Kawakami K, Dyer C. Neural Architectures for Named Entity Recognition. In: Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT); 2016. p. 260-70.

[10] Lee J, Yoon W, Kim S, Kim D, Kim S, So CH, et al. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics. 2020;36(4):1234-40.

[11] Alsentzer E, Murphy J, Boag W, Weng WH, Jindi D, Naumann T, et al. Publicly Available Clinical BERT Embeddings. In: Proceedings of the 2nd Clinical Natural Language Processing Workshop; 2019. p. 72-8.

[12] OpenAI, Achiam J, Adler S, Agarwal S, Ahmad L, Akkaya I, et al.. GPT-4 Technical Report. arXiv; 2023. Available from: https://arxiv.org/abs/2303.08774.

[13] Singhal K, Azizi S, Tu T, Mahdavi SS, Wei J, Chung HW, et al. Large language models encode clinical knowledge. Nature. 2023 Jul;620(7972):172–180. Available from: http://dx.doi.org/10.1038/s41586-023-06291-2.

[14] Brown TB, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, et al..

Language Models are Few-Shot Learners. arXiv; 2020. Available from: https://arxiv.org/abs/2005.14165.

[15] Yang X, Chen A, PourNejatian N, Shin HC, Smith KE, Parisien C, et al.. GatorTron: A Large Clinical Language Model to Unlock Patient Information from Unstructured Electronic Health Records. arXiv; 2022. Available from: https://arxiv.org/abs/2203.03540.

[16] Rohanian O, Nouriborji M, Kouchaki S, Clifton DA. On the effectiveness of compact biomedical transformers. Bioinformatics. 2023 Feb;39(3). Available from: http://dx.doi.org/10.1093/bioinformatics/btad103.

[17] Rohanian O, Nouriborji M, Jauncey H, Kouchaki S, Nooralahzadeh F, Clifton L, et al. Lightweight transformers for clinical natural language processing. Natural Language Engineering. 2024 Jan;30(5):887–914. Available from: http://dx.doi.org/10.1017/S1351324923000542.

[18] Kohane IS, Churchill SE, Murphy SN. A translational engine at the national scale: informatics for integrating biology and the bedside. Journal of the American Medical Informatics Association. 2012 Mar;19(2):181–185. Available from: http://dx.doi.org/10.1136/amiajnl-2011-000492.

[19] Radford A, Narasimhan K, Salimans T, Sutskever I. Improving Language Understanding by Generative Pre-Training. OpenAI; 2018. Available from: https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf.

[20] Karabacak M, Margetis K. Embracing Large Language Models for Med-

ical Applications: Opportunities and Challenges. Cureus. 2023 May. Available from: http://dx.doi.org/10.7759/cureus.39305.

[21] Rawte V, Sheth A, Das A. A Survey of Hallucination in Large Foundation Models. arXiv; 2023. Available from: https://arxiv.org/abs/2309.05922.

[22] Zhao WX, Zhou K, Li J, Tang T, Wang X, Hou Y, et al. A Survey of Large Language Models. arXiv preprint arXiv:230318223. 2023. Available from: https://arxiv.org/abs/2303.18223.

[23] Grattafiori A, Dubey A, Jauhri A, Pandey A, Kadian A, Al-Dahle A, et al. The Llama 3 Herd of Models. arXiv preprint arXiv:240721783. 2024. Available from: https://arxiv.org/abs/2407.21783.

[24] Brat GA, Weber GM, Gehlenborg N, Avillach P, Palmer NP, Chiovato L, et al. International electronic health record-derived COVID-19 clinical course profiles: the 4CE consortium. npj Digital Medicine. 2020 Aug;3(1). Available from: http://dx.doi.org/10.1038/s41746-020-00308-0.

[25] OpenAI. tiktoken. GitHub; 2024. GitHub repository, release 0.6.0 (published 2024-02-09). Available from: https://github.com/openai/tiktoken.

[26] Isaacus-dev. semchunk. GitHub; 2024. GitHub repository, version 2.0.0 (released 2024-06-19). Available from: https://github.com/isaacus-dev/semchunk.

[27] Liu F, Shareghi E, Meng Z, Basaldella M, Collier N. Self-alignment Pretraining for Biomedical Entity Representations. In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies; 2021. p. 4228-38.

[28] He T, Belouali A, Patricoski J, Lehmann H, Ball R, Anagnostou V, et al. Trends and opportunities in computable clinical phenotyping: A scoping review. Journal of Biomedical Informatics. 2023 Apr;140:104335. Available from: http://dx.doi.org/10.1016/j.jbi.2023.104335.

[29] Hou J, Zhao R, Gronsbell J, Lin Y, Bonzel CL, Zeng Q, et al. Generate Analysis-Ready Data for Real-world Evidence: Tutorial for Harnessing Electronic Health Records With Advanced Informatic Technologies. Journal of Medical Internet Research. 2023 May;25:e45662. Available from: http://dx.doi.org/10.2196/45662.

[30] Johnson AEW, Pollard TJ, Shen L, Lehman LwH, Feng M, Ghassemi M, et al. MIMIC-III, a freely accessible critical care database. Scientific Data. 2016 May;3(1). Available from: http://dx.doi.org/10.1038/sdata.2016.35.

[31] Feinstein AR, Cicchetti DV. High agreement but low kappa: I. The problems of two paradoxes. Journal of Clinical Epidemiology. 1990;43(6):543-9.

[32] Byrt T, Bishop J, Carlin JB. Bias, prevalence and kappa. Journal of Clinical Epidemiology. 1993;46(5):423-9.

[33] Hripcsak G, Rothschild AS. Agreement, the F-measure, and reliability in information retrieval. Journal of the American Medical Informatics Association. 2005;12(3):296-8.

[34] Uzuner Ö, South BR, Shen S, DuVall SL. 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. Journal of the American Medical Informatics Association. 2011;18(5):552-6.

[35] Norgeot B, Quer G, Beaulieu-Jones BK, Torkamani A, Dias R, Gianfrancesco M, et al. Minimum information about clinical artificial intelligence modeling: the MI-CLAIM checklist. Nature Medicine. 2020;26(9):1320-4.

[36] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 2024;385:e078378.

[37] John Snow Labs. Healthcare NLP (Spark NLP for Healthcare), version 6.3; 2026. Production-grade clinical NLP platform; academic free licence. https://www.johnsnowlabs.com/healthcare-nlp/.

[38] Lopez I, Swaminathan A, Vedula K, et al. Clinical entity augmented retrieval for clinical information extraction. npj Digital Medicine. 2025;8:45.

[39] Ntinopoulos V, et al. Large language models for data extraction from unstructured and semi-structured electronic health records: a multi-

ple model performance evaluation. BMJ Health & Care Informatics. 2025;32:e101139.